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Multi-megawatt pulses at 50 MHz from a single-pump Mamyshev oscillator gain-managed amplifier laser

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We have developed a compact all-PM-fiber ytterbium-doped Mamyshev oscillator-amplifier laser system generating compressed pulses of 102 nJ and 37 fs, thus having over 2 MW of peak power, at a repetition rate of 52 MHz. The pump power from a single diode is shared between a linear cavity oscillator and a gain-managed nonlinear amplifier. The oscillator is self-started by pump-modulation and a linearly polarized single-pulse operation is achieved without filter tuning. The cavity filters are near-zero dispersion fiber Bragg gratings with a Gaussian spectral response. To our knowledge, this simple and efficient source has the highest repetition rate and average power among all-fiber multi-megawatt femtosecond pulsed laser sources and its architecture holds potential for generating higher pulse energies.

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Ultrafast fiber laser sources are used in a wide variety of applications across life sciences, industry, and scientific areas. Typical examples of these applications are multiphoton and time-resolved microscopy, femtosecond micromachining, generation of higher harmonics, supercontinuum or terahertz waves, and two-photon polymerization. These applications usually rely on a stable source of high-energy ultrashort pulses provided by a low-power femtosecond oscillator amplified by a complex system involving several components and free-space propagation. To improve the energy of the oscillators and reduce the complexity of the amplifying system, a new ultrafast laser architecture named Mamyshev oscillator (MO) was recently introduced [1–5] and eventually developed to reach unprecedented peak power levels in ytterbium-doped systems emitting at 1060 nm [6,7]. Such high-energy oscillators were also demonstrated in erbium-doped fibers emitting at 1565 nm [8] and thulium-doped fibers emitting at 1965 nm [9]. Early demonstrations of high-energy MO were based on ring cavities involving free-space propagation sections to provide more control and facilitate the exploration of the parameters [4–10]. Linear cavities with free-space segments were also considered for their simplicity [4,11]. In addition, the benefits of an all-fiber format are obvious and different groups reported progress in that area [12–16]. Most notably, Haig *et al.* obtained 80-nJ, 40-fs compressed pulses from a self-starting all-fiber ring cavity MO based on a single 10-μm core Yb fiber segment [17] designed as a gain-managed nonlinear amplifier (GMN) [18]. One outstanding issue about MO is their capacity to self-start, i.e., to reach the mode-locked regime without the use of an external seed laser [3]. The architecture of MO does not support CW oscillation due to the offset between the two spectral filters. For an MO to start, the initial perturbation must undergo sufficient spectral broadening by self-phase modulation (SPM) to overlap both filters. In several implementations, this perturbation is provided by an external pulsed laser seed or nonlinear feedback [8,10,14]. Nonetheless, the most practical technique remains using pump modulation in combination with a reduced filter offset [5,11] although a dynamical adjustment of the filter offset must usually be done to optimize the system's performance after mode-locking is achieved.

We introduce here a monolithic, all-PM-fiber, ytterbium-doped Mamyshev oscillator gain-managed amplifier (MOGMA) laser system. The linear cavity oscillator is terminated by fiber Bragg gratings. The fiber amplifier is spliced directly on the oscillator and they both share the pump power emitted by a single diode. The oscillator is designed to ensure self-starting through pump modulation, while avoiding SBS-induced damage [11]. Its short fiber length allows for reduced energy and spectral broadening within the oscillator and favors single-pulse operation. The amplifier is designed to bring the pulses to a high energy and duration below 40 fs by taking advantage of the GMN pulse evolution [18]. This efficient, compact, and single-unit fiber laser source generates pulses centered at 1060 nm with a repetition rate of 52 MHz, an energy of 102 nJ, a FWHM duration of 37 fs, and a peak power over 2 MW after compression by a pair of gratings. The purely core-guided signal throughout the laser facilitates heat-management and paves the way for energy, peak power, and average power scaling in larger mode area fibers.

The oscillator-amplifier setup is shown in Fig. 1. The pump laser diode (BWT, K976AAHRN) can provide up to 27 W and

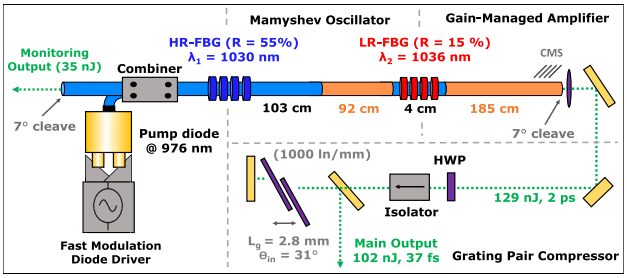


Fig. 1. MOGMA laser setup. The blue/orange sections are Coractive passive 10/125DC-PM and gain DCF-YB-10/125E-PM fibers, respectively. HR/LR-FBG, high/low reflectivity fiber Bragg gratings; CMS, cladding mode stripper; and HWP, half-wave plate.

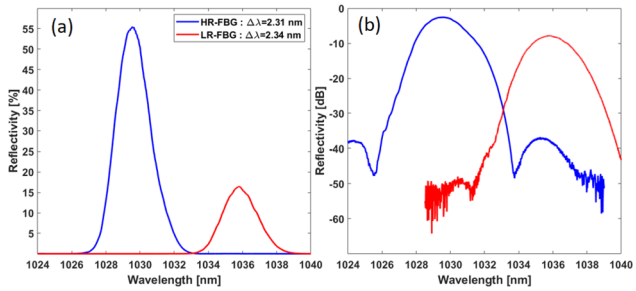


Fig. 2. Measured slow axis reflectivity profiles of the fiber Bragg gratings on (a) linear and (b) log scales.

is stabilized at 976 nm to maximize absorption. The Mamyshev oscillator has a linear configuration terminated by high- (HR-FBG) and low-reflectivity fiber Bragg gratings (LR-FBG) at 1030 nm and 1036 nm, respectively. The oscillator is followed by the GMN amplifier which is spliced directly on the LR-FBG and terminated by a cladding mode stripper to eliminate the residual pump power. The output yields chirped high-energy picosecond pulses. A standard pulse compressor based on a pair of high-quality transmission gratings (II-VI LightSmyth 1040 nm, 1000 grooves/mm, 85% overall efficiency) is then used to compress the pulses in the femtosecond regime to achieve high peak power. The whole fiber length is made of 10-μm core double-clad polarization-maintaining fibers. The length of the gain fiber within the oscillator (L_Y) was chosen experimentally to provide enough gain to self-start while keeping as much pump power as possible for the amplifier. The reduced intracavity saturated gain also prevents CW breakthrough or multi-pulsing at the operational pump power, even without filter-tuning. The length of the passive fiber near the HR-FBG was adjusted to obtain a repetition rate near 50 MHz and provides additional spectral broadening to the less energetic backward propagating pulse. A different cavity length could have been chosen to obtain a desired repetition rate (roughly 10 to 100 MHz) without fundamental modifications to the setup. Higher/lower intracavity gain would be required to self-start for higher/lower repetition rates. The passive fiber length between the oscillator and the amplifier was minimized to reduce dispersion and nonlinear perturbations in the GMN amplifier pulse evolution. Ideally, the LR-FBG would have to be inscribed directly within the gain fiber to avoid those perturbations. As discussed later, the amplifier length (L_A) was chosen to optimize the gain-managed nonlinear amplification and allow efficient compression of the high-energy pulses to femtosecond duration. Angled cleaves at both fiber outputs are critical to avoid parasitic CW lasing within the laser system. The low reflectivity of the LR-FBG is required to maintain the pulse compressibility after the amplifier. Since the pulse incident on this FBG is linearly chirped, a stronger spectral filtering at this position would result in a temporal splitting of the pulse. Nonetheless, the weakened feedback from the LR-FBG still allows for enough spectral broadening to overlap the HR-FBG in the opposite direction due to the long passive fiber segment and high spectral energy density of the incoming pulse, thus maintaining the cavity in oscillation.

The reflectivity profiles of the fiber Bragg gratings are shown in Fig. 2. The combination of large bandwidth (>2 nm) and high reflectivity was achieved by inscribing a very short grating length (approximately 120-μm FWHM Gaussian apodization) with high index modulation (Δn up to 2.5×10^{-3}). To fulfill all those requirements in a large mode area, low-NA, and thus low-photosensitivity fiber, a proprietary inscription method from TeraXion based on the phase-mask femtosecond inscription technique [19] was required. The bandwidth of the FBG was limited by how short we could produce gratings with a smooth Gaussian reflectivity profile.

Mode-locking of the oscillator is achieved by pump modulation with a 100-kHz square wave at 25% duty cycle and 20-W peak-to-peak amplitude. We used a pump driver (MESSTEC, FM 20-06) with a fast rise time of 50 ns. The 100-kHz frequency is a reliable but not critical value since we observed mode-locking with modulation frequencies ranging from 70 to 350 kHz. In our experience, keeping the average power as low as possible with a sharp modulation amplitude leads to reliable single-pulse mode-locking while avoiding damage to the fibers. The use of a lower duty cycle is thus beneficial. It is also possible to obtain multi-pulse mode-locking without pump modulation by using an extremely high pump power (20 W) and a smaller filter offset (~ 5 nm). Single-pulse operation is obtained from there by lowering the pump power, but a significant filter overlap CW does breakthrough when the pump is brought back up.

Due to the MO properties [14], the laser always starts in a linearly polarized state along a single axis of the PM-fiber. Furthermore, the high birefringence (3×10^{-4}) of the fiber implies a center wavelength shift of approximately 0.3 nm for each FBG between the slow and fast axes. To take advantage of this, the HR-FBG and the gain fiber were spliced together with their panda rods aligned at a 90° rotation angle relative to each other. This causes a 0.6-nm spectral filter offset difference between the two linearly polarized states of the cavity (HR-slow/LR-fast versus HR-fast/LR-slow). This difference results in a polarization-dependent loss which is significant enough to always self-start in the favored polarization state without any polarizer in the cavity. Otherwise, the odds of starting in a given polarization state would be near 50/50. Over one hundred tests with these settings, we found a start rate of 100% for single-pulse near-instantaneous mode-locking in the chosen polarization state. Once mode-locking is achieved, the pump modulation is stopped and the average pump power is increased up to 15 W for optimal performance.

Measured and computed characteristics of the pulse are shown in Fig. 3. Before the isolator (T= 93%), this MOGMA laser yields 129-nJ and 2-ps pulses at a repetition rate of 52 MHz for an average signal power of 6.71 W and optical laser efficiency

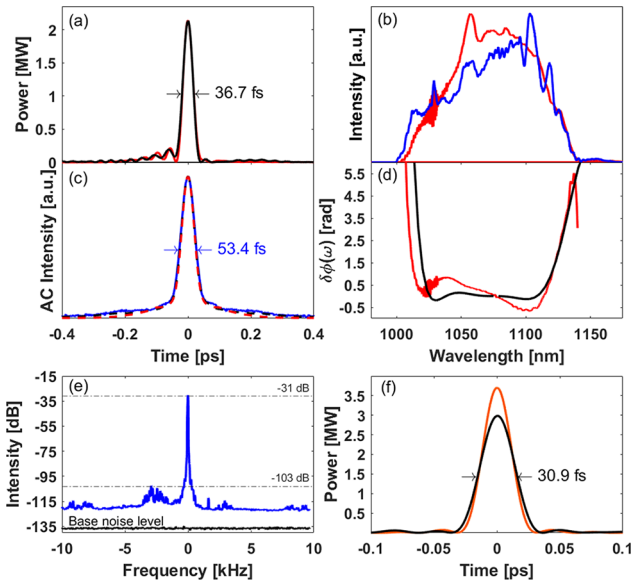


Fig. 3. (a) Temporal pulse profile, (b) linear optical spectrum, (c) autocorrelation trace, and (d) spectral phase profile for measured (blue), 100-pulse average simulated (red), and reconstructed (black) data. (e) Radio frequency (RF) spectrum. (f) Simulated pulse compressed to the fourth order (black) and transform-limited (orange).

of 45%. After the passage in the grating pair, an intensity autocorrelation trace reveals a pulse FWHM duration of 36.7 fs. The remaining average power is 5.30 W, for a pulse energy of 102 nJ. The spectrum reveals a spectral shoulder between 1140 and 1200 nm associated with stimulated Raman scattering (SRS). Our simulations are in excellent agreement with the experimental results and they confirmed this point. The Raman energy fraction is computed to be only 0.6% of the total pulse energy. For an accurate estimation of the peak power taking into account the energy lost in the sidelobes, a numerical reconstruction of the pulse is done with the PICASO algorithm [20]. We find a peak power of 2.13 MW with 78% of the energy in the central lobe. The compressed pulse spectral phase profile shows a lack of β_4 compensation from the transform-limited pulse. From the simulation, a phase correction up to the fourth order yields a near transform-limited pulse of 31 fs and 3-MW peak power. Of course, a more complex compressor might also induce more losses and would not necessarily be advantageous. The radio frequency spectrum (resolution bandwidth of 50 Hz) shows a good signal to noise ratio of 72 dB despite the high round trip gain and nonlinear phase in the system.

The numerical model is based on the well-known generalized nonlinear Schrödinger equation and accounts for dispersion up to the fourth order, self-phase modulation (SPM), SRS, and self-steepening. The wavelength-dependent gain is computed from the ytterbium absorption and emission cross sections with the population equations solved in the steady state and including the amplified spontaneous emission (ASE) [21]. The simulations allow us to analyze the intrafiber pulse dynamics and highlight a path for improvements. Within the oscillator, the short cavity length, narrow filters, and high gain yield a pulse evolution heavily dominated by nonlinearity. The parabolic attractor does not have enough propagation distance to fully smooth the spectrum [22]. As a result, the pulse reaching the LR-FBG exhibits

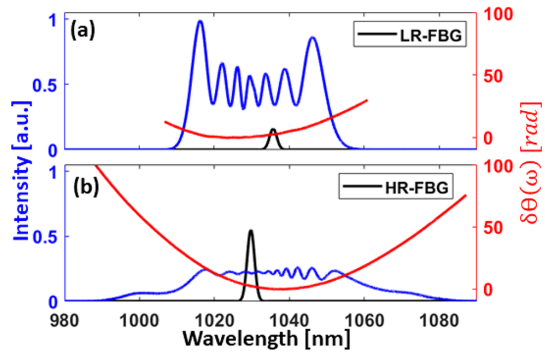


Fig. 4. Simulated pulse spectrum before reaching (a) the LR-FBG and (b) the HR-FBG within the oscillator.

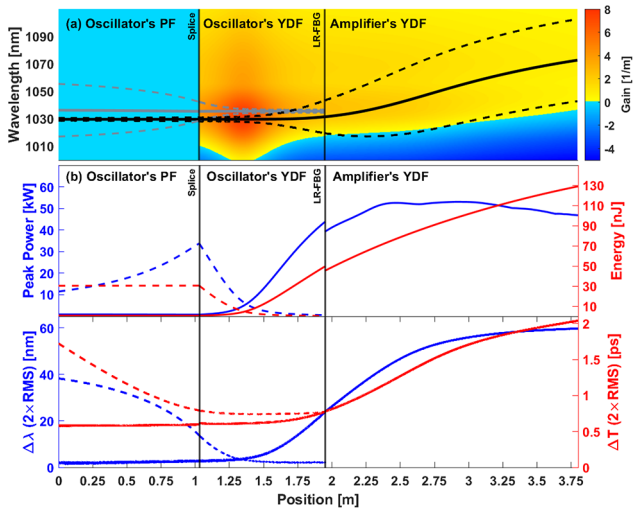


Fig. 5. (a) Mean pulse wavelength (full lines) and 2×rms spectrum width (dashed lines) relative to the gain map in the forward/backward (black/gray) directions. (b) Pulse energy, peak power, time, and spectral width evolution along the fiber length in the forward/backward (full lines/dashed lines) directions. PF/YDF, passive/Yb-doped fibers.

strong spectral modulations, as can be seen in Fig. 4, typical of SPM with moderate dispersion. In the other direction, the pulse has a high energy in the passive fiber and aggregates a much larger nonlinear phase. In that direction, dispersion plays a greater role which helps reduce the depth of spectral modulations and maintain temporal stability. The oscillator is stable for pump powers from 2 to 6 W and from 8 to 18 W.

Above 18 W, the oscillator is unstable due to excessive spectral broadening and high losses. Between 6 and 8 W, the oscillator exhibits an instability zone where the pulse cannot reach steady state. This behavior was expected and previously studied by Želudevičius *et al.* for an MO dominated by SPM [12]. This is the case here at low gain where the pulse does not spectrally broaden fast enough for dispersion to be influential in both directions. In the unstable pump region, the pulse spectral lobes never coincide with the filter windows to prevent fission. At a pump level near threshold (2 to 6 W), the LR-FBG filter stays within the first spectral lobe and a stable state is again observed.

As can be seen in Fig. 5, the pulse clearly exhibits a GMN evolution within the oscillator-amplifier as it co-propagates with the pump, starting from the 0.7-nJ and 0.7-ps Gaussian pulse

reflected by the HR-FBG. Notably, as the spectrum of the pulse broadens, its shorter wavelengths undergo absorption which induces an asymmetric spectral broadening with a significant shift of the mean spectrum toward longer wavelengths [18]. As the absorption becomes critical, the pulse deviates from the parabolic pulse attractor and we observe a saturation of the spectral broadening and peak power while the pulse energy and duration keep growing. Those effects help to slow down the onset of SRS and wave breaking. By proper management of the amplifier length, input pulse wavelength, energy, duration, and pump power, we achieved a very high output pulse energy with very short duration after compression. Previous implementations of the GMN regime using similar gain fiber parameters (10- μm core diameter and cladding absorption of 6.4 dB/m @975 nm) reported a pulse energy between 80 and 190 nJ with pulse duration ranging from 33 to 40 fs [10,17,18,23]. Here, we obtain similar performances despite the oscillator integration causing dependencies between seed energy and pump power. For the GMN amplifier, energy scaling above the 1-μJ level in a 30-μm-core fiber was reported [24]. Accordingly, we expect the MOGMA laser to follow a similar trend in terms of energy scaling. While promising, such fibers are slightly multimode which adds some interesting challenges, especially regarding the FBGs; further studies will be required to find a good configuration.

With this laser system, as with the previous studies mentioned above, the main limitation is SRS. When we try to pump harder or use a longer amplifier to get an energy above 130 nJ, SRS starts to transfer a critical portion of the pulse energy in the Raman shoulder and pulse compressibility is lost. To observe a similar SRS intensity in the simulations, it is necessary to add an amplifier input noise of 650 photons/Hz above 1100 nm in the same polarization state as the pulse. Note that our total computed ASE power is 1.4 μW at the main output, mostly around 1030 nm, and does not explain this noise figure. From simulations, the standard deviation for Raman-induced peak power fluctuations is found to be 0.7% over 100 pulses. A similar MOGMA was optimized for a low-noise scenario (1 photon/Hz) and at a repetition rate of 50 MHz (pump power = 25 W, L_P = 1.55 m, L_Y = 0.50 m, L_A = 3.0 m). Without accounting for free-space propagation losses, a 25-fs pulse of 300 nJ and up to 9-MW peak power is achieved before wave breaking with the same fiber. Therefore, further study into a strategy to reduce the emission or filter out the Raman-induced noise in the GMN amplifier would be greatly beneficial for performance and stability.

To summarize, we developed a new MOGMA laser architecture, where an SPM-dominant, pulse energy and spectral width restrained MO is combined with a GMN amplifier to achieve high energy (>100 nJ) and very short (<40 fs) pulse generation. Furthermore, we have shown that it can be implemented in a cost-efficient single-pump all-PM-fiber configuration with FBGs as filters. In this specific demonstration, we used femtosecond-inscribed low-dispersion FBGs, allowing for a shorter cavity length and a high repetition rate of 52 MHz. We also got rid of the need for an intracavity polarizer to select the polarization state by taking advantage of the FBG polarization-dependent resonance wavelength within a PM fiber. The result is a purely core-guided fiber laser compatible with a very high average power. Those features are of great interest for many power-hungry applications such as video-rate multi-photon microscopy, micro-machining, texturing, and 3D printing. In addition, we anticipate further power scaling based on the use of larger mode area fibers. An all-fiber MOGMA laser generating >1 μJ sub-40-fs pulses with >100 W of average power might be on the radar. Such a powerful source could easily be combined with nonlinear fibers or crystals to generate multi-watt power laser light at various wavelengths. We emphasize that the MOGMA laser can be tailored for specific applications at lower or higher repetition rates or energies to provide sub-100-fs pulses. Of course, the GMN amplifier approach is restrained by the nonlinear intensive pulse evolution process inducing many dependencies between pulse energy, time duration, and spectral width. For a very high spectral energy density, a chirped-pulse amplification system (CPA) is still the favored method. However, the MOGMA architecture is super-robust, compact, cost-efficient, and can significantly overcome the gain-narrowing limit at high average power. It is expected to be an elegant solution for a variety of applications.

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Data availability. Data underlying the results presented in this paper may be obtained from the authors upon request.

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